Project Initialization and Planning Phase

Date	29 June 2026
Team ID	
Project Title	A comprehensive measure of well being
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to make human development analysis fast and data-driven using machine learning, replacing slow manual HDI calculation with an instant, explainable prediction tool. It tackles the inefficiency of comparing countries by hand, promising faster insight, transparent sub-index breakdowns, and better-informed policy decisions. Key features include an ML-based HDI classification model and instant sub-index visualization.

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Project Overview	
Objective	The primary objective is to make human development assessment fast and data-driven by using machine learning to instantly predict a country's HDI tier from its core indicators.
Scope	The project covers data ingestion, sub-index computation, ML-based tier classification, visualization, and report generation for single countries or batch comparisons.
Problem Statement	
Description	Manually collecting UNDP/World Bank indicators and computing the HDI score is slow, error-prone, and gives no insight into which sub-index is weakest.
Impact	Solving this will let researchers and policymakers get instant, explainable HDI predictions, speeding up analysis and enabling faster, better-targeted development decisions.
Proposed Solution	
Approach	Employ a trained classification/regression model to compute Health, Education, and Income sub-indices from raw indicators, combine them into an overall HDI score, and classify the result into Very High / High / Medium / Low tiers.
Key Features	- ML-based HDI sub-index and tier prediction model.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard CPU (no GPU required)
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	20 GB SSD
Software		
Frameworks	Python frameworks	FastAPI
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, VS Code
Data		
Data	Source, size, format	UNDP Human Development Report indicators, ~195 countries, csv